

PROJECT
TITLE

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GitHub: <http://www.github.com>
Visualization/Demo: <http://www.demo.com>

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1 Project Idea and Use Case

1.1 Topic Introduction

Here you introduce the topic, give an overview of what you're going to discuss, and why the topic is important.

Examples of how to add references using BibTex are presented here: [Knuth, 1984; 1986; Lamport, 1994; Lesk and Kernighan, 1977; Mittelbach *et al.*, 2004](#). You

1.2 Motivation

Explain why you have chosen this project.

1.3 Problem Definition and Goals

1.3.1 Business and Research Questions

Present the questions that your data lake implementation will answer.

1.3.2 Business Intelligence Questions

Present the questions that your data warehouse implementation will answer.

2 Data Lake

2.1 Data Sources

Explain why the data sources support your use case (data lake). Describe the data sources. Analyze the data sources for potential limitations.

- Explain why the data sources support your use case (data lake)
- Describe the data sources. Analyze the data sources for potential limitations.
- Data catalog (Glue/Data Catalog) with tags, owners, and descriptions.
- Schema/versioning strategy; schema evolution policy.
- Lineage tracking (within orchestrator/catalog if possible).

2.2 Architecture

Describe the system setup and all elements needed to answer your business questions (data lake).

- Cloud infrastructure
- Storage layers & zones
- File formats
- Catalog service
- Core compute engines
- Streaming vs batch ingestion patterns
- Interop with data warehouse/data lake

3 Data Ingestion

3.1 Data Source Documentation

- Clear identification of each source (dynamic vs static).
- API details: authentication, rate limits, pagination handling, endpoints.
- Justification of why the source supports the use case.

3.2 Pipeline Design

- Architecture of the ingestion process (batch, hybrid).
- Tools/technologies used (e.g., Glue, Lambda, etc).
- Diagrams representing the ingestion workflow.

3.3 Data Capture & Handling

- Handling of schema evolution (e.g., column changes).
- Strategies for retries, error handling, and logging.
- Management of duplicates or late-arriving data.

4 Data Storage

4.1 Storage Layers & Services

- Technologies chosen (S3, relational DBs, vector DBs, etc).

4.2 Data Formats & Optimization

- Choice of file formats (CSV, Parquet, JSON) with reasoning.

4.3 Database Design

- For relational: ER diagrams, keys, and relationships.
- For NoSQL/vector: indexing, sharding, retrieval strategies

5 Data Transformation

5.1 Transformation Logic

- Step-by-step description of transformations (joins, aggregations, filtering, enrichment).
- Flow diagrams showing transformation pipelines.

5.2 Data Cleaning & Preparation

- Techniques for handling missing values, outliers, duplicates.
- Assumptions made in imputation or filtering.
- Standardization (units, formats, types).

5.3 Data Quality & Validation

- Analysis for accuracy, consistency, completeness, timeliness.
- Monitoring of data drift or anomalies

5.4 Scalability & Reusability

- Reusability of transformation scripts.
- Modularity of pipelines.

6 Data Warehouse

6.1 Business goals and objectives

- Identify key stakeholders and decision-makers.
- Understand required KPIs, metrics, and reports.
- Determine data granularity (daily, monthly, transactional). iz

6.2 Data Warehouse Modeling

Present the model used and implementation of your data warehouse.

6.3 Architecture

Describe the system setup and all elements needed to answer your business questions (data lake).

6.4 Data Preparation

Explain why the data sources support your use case (data warehouse). Describe the data sources. Describe the ETL/ELP process required.

- Design an extraction strategy from source systems.
- Define transformation rules (cleansing, standardization, enrichment).
- Implement loading strategy (incremental vs full load).

7 Data Visualization

7.1 Visualization Purpose

- What problem are you solving?
- Who is the target audience?
- What decisions should the visualization help with?

7.2 Theoretical Model

Describe the theoretical framework for the implementation of visualizations.

7.3 Data Warehouse Visualization

Present visualizations, based on the theoretical framework, how the data warehouse answers the business intelligence questions proposed in section [1.3.2](#).

7.4 Answer to Business Questions

Provide answers to your proposed business questions using visualizations.

8 Conclusions

This section does not need to be long, but it needs to be concrete. Explain what you found, what you did not find, and what needs to be done next. No more than two pages maximum.

8.1 Discussion of the Solution

Proper discussion of the solution. Highlight the advantages, disadvantages, trade-offs in terms or technologies used etc.

8.2 Project Outcomes

Proper discussions of the project outcomes.

8.3 Future Work

Areas to improve the project. What would you do different?

8.4 GenAI Declaration

Describe how and where you use GenAI solutions.

Bibliography

- Knuth, D. E. (1984). Literate programming. *The Computer Journal*, 27(2), 97–111.
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- Lamport, L. (1994). *ΛT_EX: A document preparation system* (2nd ed.). Addison Wesley.
- Lesk, M., & Kernighan, B. (1977). Computer typesetting of technical journals on UNIX. *Proceedings of American Federation of Information Processing Societies: 1977 National Computer Conference*, 879–888.
- Mittelbach, F., Gossens, M., Braams, J., Carlisle, D., & Rowley, C. (2004). *The ΛT_EX companion* (2nd ed.). Addison-Wesley Professional.

B Appendix - if Needed

The first appendix is here. The purpose of appendices is to keep the main part of the document clear and readable. There can be many appendices, everyone should form an integral part. The following should be considered as the candidates to build appendices:

- Multi-page tables disturbing the reading with important information which you want to include in your manuscript;
- detailed schematics for which you provided a simplified version in text (you can refer to the more complex scheme in an Appendix and only show the simplified scheme inside the main text).
- important pieces of code allowing for validation of the results. Most of the code will probably go to the git repository anyway.
- multiple figures of some properties, showing that the solution described in the manuscript is capable of calculating the properties shown in an Appendix, but the values in figures do not have the crucial significance for the research/engineering plot line of your manuscript.

C Appendix - Project Evaluation Grid

Table C.1. Final Report Evaluation Grid

Evaluation Criteria	Maximum possible points	Student 1	Student 2	Student 3
1. Introduction	5			
• Topic introduction • Motivation • Problem definition & Goals • Business/Research questions • Business intelligence questions			X	
2. Data Lake	5			
• Data Assumptions • Architecture		X		
3. Data Ingestion (Data Lake & Data Warehouse)	5			
• Data Source Documentation (dynamic/static, API details, justification) • Pipeline Design (batch/streaming, workflow diagram, tools) • Data Capture & Handling (schema evolution, error handling, duplicates)			X	
4. Data Storage	5			
• Storage Layers & Services (zones, technologies) • Data Formats & Optimization (formats, partitioning, compression) • Database Design (ER diagrams, indexing, retrieval)		X		
5. Data Transformation (Data Lake & Data Warehouse)	5			
• Transformation Logic (joins, aggregations, enrichment) • Data Cleaning & Preparation (missing values, assumptions) • Data Quality & Validation • Scalability & Reusability (modular, reusable pipelines)		X	X	
6. Data Warehouse	5			
• Define business goals and objectives • Data modeling approach • Architecture • Data preparation			X	
7. Data Visualization	5			
• Visualization purpose • Theoretical model • Data Warehouse visualizations • Answer business questions			X	
8. Conclusions	5		X	
• Discussion of the solution • Discussions of the project outcomes • Future work • GenAI Declaration	5	X	X	X

D **Appendix - Project Code Leads**

Table D.1. Code Project Leads

Responsibilities	Student 1	Student 2	Student 3
API Data Source 1 (Ingestion)	X		
API Data Source 2 (Ingestion)		X	
Additional Data Sources (Ingestion)			X
Databases or S3 services	X	X	X
Any other scripts	X	X	
Jupyter Notebooks (Pipeline - Design)		X	
...		X	